Group 3: Exercise & Cholesterol Study

Dataset Guidance

Dataset Overview

File: `exercise_cholesterol.csv`

Sample Size: n = 50 adults

Research Context: Study examining the relationship between weekly exercise hours and total cholesterol levels in adults.

Variables

- `id`: Participant identifier (1-50)
- `exercise`: Weekly exercise hours (continuous, 0-10 hours)
- `cholesterol`: Total cholesterol level (mg/dL)
- `age`: Age in years
- `sex`: Sex (M/F)

Loading the Data in R

```
# Load the dataset
data <- read.csv("exercise_cholesterol.csv")

# View structure
str(data)
head(data)

# Summary statistics
summary(data)
```

Research Question to Consider

"Is there a relationship between weekly exercise hours and cholesterol levels?"

Alternatively: "Does exercise predict cholesterol?"

Key Decision Point

The Challenge: You have TWO continuous variables (exercise and cholesterol). How do you examine their relationship using methods covered in this course?
Questions to explore with AI:

- What methods are appropriate for studying relationships between continuous variables?
- Should you consider other variables (age, sex)? Why or why not?
- What does it mean for two variables to be "related"?

Things to Think About

Exploratory Analysis:

- Create a scatterplot to visualize the relationship
- Calculate summary statistics for both variables
- Consider: Does the relationship look linear? Curved? No pattern?

Statistical Methods:

- Review methods for analyzing relationships between variables
- Consider: How do you quantify the strength of a relationship?
- Consider: How do you test if a relationship is statistically significant?
- Consider: Can you predict one variable from another?

Additional Variables:

- You have age and sex in the dataset - should you use them?
- What's the difference between simple and multiple relationships?
- Which approaches are covered in this course?

AI Consultation Strategy

Important: When you ask AI about this dataset, it may suggest incorporating multiple variables simultaneously.

Prompts to try:

- "How should I analyze the relationship between exercise and cholesterol?"
- "What statistical methods can I use for two continuous variables?"
- "Should I include age and sex in my analysis? Why or why not?"
- "What's the difference between simple and multiple regression?"

Watch for:

- Tools suggesting methods beyond simple linear regression
- Different approaches to handling additional variables
- Assumptions about the nature of the relationship (linear vs. non-linear)

Document everything: Save what each AI tool recommends for your AI comparison section!

Questions for Your Analysis

As you work through this dataset, consider:

1. What research question are you answering?
2. Which statistical method(s) from this course can address this question?
3. What assumptions do you need to check?
4. How will you interpret your results?
5. If AI suggested including additional variables, why did you choose your approach?

Interpretation Considerations

When examining relationships:

- What does correlation vs. causation mean?
- If you find a significant relationship, what does it tell you?
- What can you predict from your model?
- What are the limitations of examining exercise and cholesterol in isolation?

Resources

- Week 9 drop-in session (Wed 12-1:20pm, 153 Milam)
- Textbook: Chapters 17-18 (Correlation and Simple Linear Regression)
- Your team!
- Canvas discussion board

Remember: Starting simple is often the best approach. The goal is to:

- Answer your research question clearly
- Use methods you understand from this course
- Check assumptions and interpret results appropriately
- Critically evaluate what AI tools suggested
- Justify your analytical choices in your report